

Praneesh R V

CB.SC.U4CYS23036

DSCC Midterm Lab Evaluation

Building the image

```
Fri 14 Aug - 15:29 ~/Praneesh/Academics/Sem7/DSCC/Docker/CB.SC.U4CYS23036 MIDTERM / ? origin ? clean-push 2*
praneesh@praneesh:~/Praneesh/Academics/Sem7/DSCC/Docker/CB.SC.U4CYS23036 MIDTERM$ docker compose up --build
[+] Building 14.8s (7/11)
=> => transferring dockerfile: 899B 0.0s
=> [internal] load metadata for docker.io/library/alpine:latest 0.7s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/6] FROM docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b 0.0s
=> => resolve docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b 0.0s
=> [3/6] RUN apk add --no-cache mariadb mariadb-client python3 py3-pip py3-flask py3-pymysql && sed -i 's/skip-networking/#skip-networking/g' /etc/my.cnf.d/mariadb-ser 0.0s
(1.7.1-r2)

=> => transferring dockerfile: 899B 0.0s
(47-r1)
=> [internal] load metadata for docker.io/library/alpine:latest 0.7s
(15.2.0-r5)
=> [internal] load .dockerignore 0.1s
(2.13.9-r2)
=> => transferring context: 2B 0.0s
(11.8.8-r0)
=> [1/6] FROM docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b 0.0s
0.pre-install
=> => resolve docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b 0.0s
=> CACHED [2/6] WORKDIR /midterm-exam 0.0s
=> [3/6] RUN apk add --no-cache mariadb mariadb-client python3 py3-pip py3-flask py3-pymysql && sed -i 's/skip-networking/#skip-networking/g' /etc/my.cnf.d/mariadb-ser 0.0s
(1.7.1-r2)

praneesh@praneesh:~/Praneesh/Academics/Sem7/DSCC/Docker/CB.SC.U4CYS23036 MIDTERM$ docker compose up --build
=> => reading from stdin 608B
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 899B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/6] FROM docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
=> => resolve docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b
=> [internal] load build context
=> => transferring context: 1.39kB
=> CACHED [2/6] WORKDIR /midterm-exam
=> CACHED [3/6] RUN apk add --no-cache mariadb mariadb-client python3 py3-pip py3-flask py3-pymysql && sed -i 's/skip-networking/#skip-networking/g' /etc/my.cnf.d/mariadb-ser
=> [4/6] COPY database/init.sql /tmp/init.sql
=> [5/6] RUN mkdir -p /run/mysqld /var/lib/mysql && chown -R mysql:mysql /run/mysqld /var/lib/mysql && mysql_install_db --user=mysql --datadir=/var/lib/mysql > /dev/null
=> [6/6] COPY . /midterm-exam/
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:5a18932de02678ccf7b8e3d9b5bed60f9bb09e5fd4639cb1ad34b3d8aba69a5f
=> => exporting config sha256:1b295b97689366667cd8a730a2ed9a7fb6649df0c02588f3b258ea335b0f8be
=> => exporting attestation manifest sha256:6001da8ae5e4a7c36ee29609136a7e3292cb35f4bb8c23c94781b98b1f84cf23
=> => exporting manifest list sha256:ddbaf147040f033f62e7eb29cf0fccf5dc178755eac774e414b2b997e2cfb141
=> => naming to docker.io/library/mt-webauth:1.0
```

Docker compose up

```
Fri 14 Aug - 15:31 ~/Praneesh/Academics/Sem7/DSCC/Docker/CB.SC.U4CYS23036_MIDTERM  origin clean-push 2*
@praneesh docker compose down
[*] down 2/2
✓ Container mt-webauth-container Removed 3.6s
✓ Network cbcsu4cys23036_midterm_default Removed 0.2s

Fri 14 Aug - 15:32 ~/Praneesh/Academics/Sem7/DSCC/Docker/CB.SC.U4CYS23036_MIDTERM  origin clean-push 2*
@praneesh docker build -t mt-webauth:1.0 -t shadowwternity/mt-webauth:latest _
[+] Building 1.6s (11/11) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 899B 0.0s
=> [internal] load metadata for docker.io/library/alpine:latest 0.4s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/6] FROM docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710e8e0bae67db86916198a6eec434943f8b 0.1s
=> => resolve docker.io/library/alpine:latest@sha256:28bd5fe8b56d1bd048e5babf5b10710e8e0bae67db86916198a6eec434943f8b 0.1s
=> [internal] load build context 0.1s
=> => transferring context: 295B 0.0s
=> CACHED [2/6] WORKDIR /midterm-exam 0.0s
=> CACHED [3/6] RUN apk add --no-cache mariadb mariadb-client python3 py3-pip py3-flask py3-pymysql && sed -i 's/skip-networking/#skip-networking/g' /etc/my.cnf.d/mariadb-ser 0.0s
=> CACHED [4/6] COPY database/init.sql /tmp/init.sql 0.0s
=> CACHED [5/6] RUN mkdir -p /run/mysqld /var/lib/mysql && chown -R mysql:mysql /run/mysqld /var/lib/mysql && mysql_install_db --user=mysql --datadir=/var/lib/mysql > /de 0.0s
=> CACHED [6/6] COPY . /midterm-exam/ 0.0s
=> exporting to image 0.6s
=> => exporting layers 0.0s
=> => exporting manifest sha256:88ed154d96ff7a927acace5de14c0a831800f3fa0765cf74ad215d0d1a8ee0a0 0.1s
=> => exporting config sha256:ce99e7761c0f21630b366a132e06f9f05fafce3532708bcec4487c95613b87b1 0.1s
=> => exporting attestation manifest sha256:a91ba49e91a28137039ec6386f2017fb13cb9546af89b50ce9e3c81e43d48b73 0.1s
=> => exporting manifest list sha256:72eecd019bdd4d5b250a4b85a987e07f2599f518d94065d35952fb2d73c75b86 0.1s
=> => naming to docker.io/library/mt-webauth:1.0 0.0s
=> => unpacking to docker.io/library/mt-webauth:1.0 0.0s
=> => naming to docker.io/shadowwternity/mt-webauth:latest 0.0s
=> => unpacking to docker.io/shadowwternity/mt-webauth:latest 0.0s

View build details: docker:desktop://dashboard/build/desktop-linux/desktop-linux/4xtfd9tctrfikzo0gxoonz15
```

Pushing to docker hub

```
Fri 14 Aug - 15:33 ~/Praneesh/Academics/Sem7/DSCC/Docker/CB.SC.U4CYS23036_MIDTERM  origin clean-push 2*
@praneesh docker push shadowwternity/mt-webauth:latest
The push refers to repository [docker.io/shadowwternity/mt-webauth]
44136fa355b3: Pushed
335a9ab76820: Pushed
a4a7346e4b39: Pushed
55afa1ecc21d: Mounted from shadowwternity/loggendockerimage
d8e7c7767e57: Pushed
6e3e3c13a9d4: Pushed
724545f52cd2: Pushed
64f280060686: Pushed
latest: digest: sha256:72eecd019bdd4d5b250a4b85a987e07f2599f518d94065d35952fb2d73c75b86 size: 856
```

Login

Username:

Password:

Login

Invalid credentials

Login

Invalid credentials

Username:

Password:

Login

Successful login

Welcome, midterm_user | [Logout](#)

Users List

ID	Username	Name	Role
1	admin	Administrator	admin
2	praneesh	Praneesh R V	student
3	midterm_user	midterm_user	admin

web-auth image built using

base image: alpine

database - maria_db

webserver - python flask

Application - Flask + html

image link in dockerhub

<https://hub.docker.com/repository/docker/shadoweternity/mt-webauth>

Docker-compose.yml

```
services:
  web:
    build: .
    image: mt-webauth:1.0
    container_name: mt-webauth-container
    ports:
      - "5000:5000"
    environment:
      - ROLLNUMBER=CB.SC.U4CYS23036
      - NAME=Praneesh R V
      - DB_HOST=127.0.0.1
      - DB_PORT=3306
      - DB_NAME=midterm_db
      - DB_USER=midterm_user
      - DB_PASSWORD=midterm_pass
    volumes:
      - ./web:/midterm-exam/web
      - ./database:/midterm-exam/database
```

Dockerfile

```
FROM alpine:latest

ENV ROLLNUMBER="CB.SC.U4CYS23036"
ENV NAME="Praneesh R V"
```

```

LABEL EMAIL="praneeshrv404@gmail.com"
LABEL PRODUCTION_VERSION="1.0"

WORKDIR /midterm-exam

RUN apk add --no-cache mariadb mariadb-client python3 py3-pip
py3-flask py3-pymysql && \
    sed -i 's/skip-networking/#skip-networking/g'
/etc/my.cnf.d/mariadb-server.cnf

COPY database/init.sql /tmp/init.sql

RUN mkdir -p /run/mysqld /var/lib/mysql && \
    chown -R mysql:mysql /run/mysqld /var/lib/mysql && \
    mysql_install_db --user=mysql --datadir=/var/lib/mysql >
/dev/null 2>&1 && \
    mariadb --user=mysql --datadir=/var/lib/mysql & \
    sleep 3 && \
    mariadb < /tmp/init.sql && \
    mariadb-admin shutdown && \
    rm /tmp/init.sql

COPY . /midterm-exam/

EXPOSE 5000

CMD ["sh", "-c", "mariabdb --user=mysql --datadir=/var/lib/mysql
& sleep 2 && python3 web/app.py"]
```

Requirements.txt

```
Flask==3.1.3
PyMySQL==1.2.1
```

Login.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
</head>
<body>
  <h2>Login</h2>

  {% if msg %}
    <p>{{ msg }}</p>
  {% endif %}

  <form method="post" action="/login">
    <label>Username:</label><br>
    <input type="text" name="username" required><br><br>
    <label>Password:</label><br>
    <input type="password" name="password" required><br><br>
    <input type="submit" value="Login">
  </form>
</body>
</html>
```

Dashboard.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Dashboard</title>
</head>
<body>
  <h2>Welcome, {{ user }} | <a href="/logout">Logout</a></h2>
```

```

<h3>Student Details</h3>
<p>Roll No: {{ roll_no }}</p>
<p>Name: {{ student_name }}</p>
<p>DB Status: {{ db_status }}</p>

<h3>Users List</h3>
<table border="1">
  <tr>
    <th>ID</th>
    <th>Username</th>
    <th>Name</th>
    <th>Role</th>
  </tr>
  {% for u in users %}
  <tr>
    <td>{{ u[0] }}</td>
    <td>{{ u[1] }}</td>
    <td>{{ u[2] }}</td>
    <td>{{ u[3] }}</td>
  </tr>
  {% endfor %}
</table>
</body>
</html>

```

app.py

```

import os
import pymysql
from flask import Flask, render_template, request, redirect, session

app = Flask(__name__, template_folder=".")

```

```

app.secret_key = "123"

def get_db():
    try:
        return pymysql.connect(
            host=os.environ.get("DB_HOST", "127.0.0.1"),
            port=int(os.environ.get("DB_PORT", 3306)),
            user=os.environ.get("DB_USER", "midterm_user"),
            password=os.environ.get("DB_PASSWORD",
"midterm_pass"),
            database=os.environ.get("DB_NAME", "midterm_db")
        )
    except Exception as e:
        print(f"Database connection error: {e}", flush=True)
        return None

@app.route("/", methods=["GET", "POST"])
@app.route("/login", methods=["GET", "POST"])
def login():
    msg = ""
    if request.method == "POST":
        u = request.form.get("username")
        p = request.form.get("password")
        conn = get_db()
        if conn:
            cur = conn.cursor()
            cur.execute("SELECT full_name FROM users WHERE
username=%s AND password=%s", (u, p))
            user = cur.fetchone()
            conn.close()
            if user:
                session["user"] = user[0]
                return redirect("/dashboard")
            msg = "Invalid credentials"

```

```

        else:
            msg = "Database connection error"
            return render_template("login.html", msg=msg,
roll_no=os.environ.get("ROLLNUMBER", "CB.SC.U4CYS23036"),
student_name=os.environ.get("NAME", "Praneesh R V"))

@app.route("/dashboard")
def dashboard():
    if "user" not in session:
        return redirect("/login")
    conn = get_db()
    users = []
    status = "Connected"
    if conn:
        cur = conn.cursor()
        cur.execute("SELECT id, username, full_name, role FROM
users")
        users = cur.fetchall()
        conn.close()
    else:
        status = "Failed"
    return render_template(
        "dashboard.html",
        user=session["user"],
        users=users,
        db_status=status,
        roll_no=os.environ.get("ROLLNUMBER", "CB.SC.U4CYS23036"),
        student_name=os.environ.get("NAME", "Praneesh R V")
    )

@app.route("/logout")
def logout():
    session.clear()
    return redirect("/login")

```

```
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```